Test 2

- As usual explain everything fully using complete sentences.
- Unless mentioned otherwise, all vector spaces are finite-dimensional over an arbitrary field F.
- You may use any definitions and results stated and proved in class (clearly specify what result you are using), but no other definitions or results. In particular, you may not use terms or results developed exclusively in tutorials.

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1. Let $M = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ with entries in a field F (where 2 means $1+1$).

   (a) Find a basis for the 1-eigenspace of M. Does your answer depend on F?
   (b) For $F = \mathbb{Q}$, find an invertible matrix P with entries in $\mathbb{Q}$ such that $P^{-1}MP$ is diagonal matrix.
   (c) Now let $F = \mathbb{F}_p$ for some prime p. Find all primes p for which M is NOT diagonalizable. If you reason properly, you do not need to do any calculations here.

2. Solve the following two independent questions about inner product spaces.
   (a) Consider $\mathbb{C}^4$ equipped with the standard Hermitian inner product $\langle v, w\rangle = v^* w$. (Recall that $*$ means taking complex conjugate of each entry and transposing the matrix.) Let

$$v = \begin{pmatrix} 1 \\ i \\ 1 \\ i \end{pmatrix} \quad \text{and} \quad u = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}.$$

   Find the following three numbers: (i) A complex number λ such that $u - \lambda v$ is orthogonal to v. (ii) A positive real number c such that cv is part of an orthonormal basis for $\mathbb{C}^4$. (iii) The coordinate of u with respect to cv.

   (b) Consider the following basis of $\mathbb{R}^3$: $v_1 = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$, $v_2 = \begin{pmatrix} 0 \\ 3 \\ 3 \end{pmatrix}$, and $v_3 = \begin{pmatrix} 9 \\ 0 \\ 9 \end{pmatrix}$. Do not check that this is a basis, but use it to find an orthonormal basis $\{e_1, e_2, e_3\}$ by applying the Gram-Schmidt procedure. (As a check, you should get $e_3 = \frac{1}{3}\begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}$. However, the credit is entirely for the procedure.)

3. (a) Find a formula for the orthogonal reflection in the plane $x = 0$ in $\mathbb{R}^3$. Write your answer as $f(v) = Av$ where A is a suitable matrix.

   (b) Find a formula for the orthogonal reflection in the plane in $\mathbb{R}^3$ that is defined by the equation $2x + 3y + 6z = 0$. Write your answer as $f(v) = Bv$ where B is a product of three matrices where each of the matrices has its nine scalars explicitly listed. Justify fully! Hint: You may use the vectors below. Consider them carefully!

$$v_1 = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 3 \\ -6 \\ 2 \end{pmatrix}, \quad \text{and} \quad v_3 = \begin{pmatrix} 6 \\ 2 \\ -3 \end{pmatrix}.$$

4. Show that there is exactly one square matrix J in Jordan form with the following properties. Clearly explain your full reasoning showing how you deduce the size of the matrix and the precise block structure from the given data.

$$\operatorname{rank}(J) = 5 \qquad \operatorname{nullity}(J) = 4 \qquad \operatorname{nullity}(J^2) = 7 \qquad \operatorname{nullity}(J^3) = 8$$

5. Recall that matrices M and N with entries in a field F are called similar if there is a matrix P with entries in F such that $P^{-1}MP = N$, and that similarity is an equivalence relation.

   Let $F = \mathbb{C}$. Consider the set S of all 3×3 matrices with entries in F defined by

   $$S = \{M \mid M_{11}+M_{22}+M_{33} = 0,\ M \text{ has 10 as an eigenvalue, } M \text{ is not similar to any diagonal matrix}\}.$$

   (a) Show that M is invertible.
   (b) Show that the set S is a union of exactly two equivalence classes under similarity.
   (c) Is (a) true if F is the field of real numbers?
   (d) Is (b) true if F is the field of real numbers?

6. Miscellaneous short problems. In case you could not do something above, you may substitute in part by doing the independent problems below.

   (a) Suppose the characteristic polynomial of an operator T on a finite dimensional vector space V factors completely. Show that V has a basis with respect to which the matrix of T is upper triangular. Your proof should be formulated in terms of *operators* as much as possible.

   (b) For $n \times n$ matrices A and B, Show that if either is invertible, then the characteristic polynomial of AB is the same as that of BA. (Do at home: can you do this without the assumption about invertibility?) Is it necessary that if AB is diagonalizable, then so is BA?